

$T = 375 \text{ N}$ A1 [4]

4 (a) (i)1 amplitude = $0.4(0) \text{ mm}$ A1

(i)2 wavelength = $7.5 \times 10^{-2} \text{ m}$
(1 sig. fig. -1 unless already penalised) A1

(i)3 period = 0.225 ms C1
frequency = $1/T = 4400 \text{ Hz}$ A1

(i)4 $v = f\lambda$
 $= 4400 \times 7.5 \times 10^{-2}$ C1
 $= 330 \text{ m s}^{-1}$ A1

[6]

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(a)	(ii)	reasonable shape, same amplitude and wavelength doubled	B1	[1]
	(b)	(i)	1.7(2) μm	A1
		(ii)	$d \sin \theta = n\lambda$ (double slit formula scores 0/2)	
			$1.72 \times 10^{-6} \times \sin \theta = 590 \times 10^{-9}$	C1
			$\theta = 20.1^\circ$ (allow 20°)	A1
		(iii)	$\frac{1}{2}L = 1.5 \tan 20.1$	C1
			$L = 1.1 \text{ m}$	A1
				[5]
5	(a)	(i)	arrow from B towards A	B1